Deep Learning Project
Implementation Analysis & Fix Report

Course: Deep Learning & Neural Networks – Fall 2026
Helwan National University – Faculty of Computer Science & AI
Project: Multi-Modal Restaurant Quality Prediction

May 13, 2026

Executive Summary

This report documents the comprehensive analysis and remediation of the Multi-Modal Restaurant Quality Prediction project for the Deep Learning course at Helwan National University. The project implements five distinct deep learning architectures (DNN, CNN with Transfer Learning, LSTM, Transformer, and Multi-Modal Ensemble) on a restaurant dataset combining tabular, text, and image data.

An initial analysis identified several critical issues including data leakage, missing positional encoding in the Transformer, insufficient dataset variety, and absence of comparative analysis. All identified issues have been addressed and fixed in the current implementation.

1 Project Overview

1.1 Selected Domain

The project focuses on restaurant quality prediction using multi-modal data:

- **Tabular Data**: Price range, cuisine type, neighborhood, review counts
- **Text Data**: Customer reviews for sentiment analysis (positive/neutral/negative)
- **Image Data**: Restaurant/food images for visual quality classification

1.2 Dataset Statistics

- **Restaurants**: 150 unique establishments
- **Review Templates**: 36 unique texts (12 per sentiment class) – **expanded from 15**
- **Image Sources**: 40+ unique Unsplash URLs – **expanded from 19**
- **Features**: Tabular (4 features: price, cuisine, neighborhood, review count), text (reviews), images (224x224x3)

2 Implemented Models

1. **DNN (Deep Neural Network)**: 3 hidden layers (64, 32, 16) with dropout – predicts restaurant rating from tabular data
2. **CNN (MobileNetV2)**: Pre-trained on ImageNet, transfer learning – binary classification (high-tier vs low-tier) from images
3. **LSTM (Bidirectional)**: Two bidirectional LSTM layers with embedding – 3-class sentiment analysis from reviews
4. **Transformer**: Custom encoder with MultiHeadAttention + sinusoidal positional encoding – 3-class sentiment analysis
5. **Multi-Modal Ensemble**: CNN (MobileNetV2) + DNN branches fused via concatenation – rating prediction from images + tabular data

3 Critical Issues Found and Fixed

3.1 Fix 1: Data Leakage (CRITICAL)

Issue: The preprocessing pipeline included `rating_scaled` (StandardScaler-transformed rating) as a tabular input feature. The DNN and Multi-Modal models used this to predict `rating` as the target, essentially learning to recover the target from a scaled version of itself.

Fix: Removed `rating_scaled` from tabular features. The current feature set is: `price_encoded`, `cuisine_encoded`, `neighborhood_encoded`, `review_count_scaled`.

3.2 Fix 2: Missing Positional Encoding (CRITICAL)

Issue: The Transformer model lacked positional encoding, which is essential for the self-attention mechanism to understand word order. Without it, the model treats input as a bag-of-words.

Fix: Added sinusoidal positional encoding (`PositionalEncoding` class) between the embedding layer and the Transformer encoder, as described in "Attention Is All You Need" (Vaswani et al., 2017).

3.3 Fix 3: Insufficient Dataset Variety (MAJOR)

Issue: Only 15 review templates (5 per sentiment class) for 150 restaurants. Only 19 unique images recycled across all restaurants.

Fix: Expanded to 36 review templates (12 per sentiment class) and 40+ unique image URLs.

3.4 Fix 4: Missing Comparative Analysis (MAJOR)

Issue: No comparison between the 5 models, no ablation studies, no side-by-side evaluation.

Fix: Created `08_Model_Comparison.ipynb` that loads all models, evaluates them on test data, and generates a comparison table and bar charts.

3.5 Fix 5: Missing Architecture Justifications (MAJOR)

Issue: Architectural decisions (layer sizes, dropout rates, optimizers) were not explained.

Fix: Added detailed architecture justifications to all 5 model notebooks explaining the reasoning behind each design choice.

3.6 Fix 6: Missing Confusion Matrices (MINOR)

Issue: CNN, LSTM, and Transformer imported `confusion_matrix` but never used it.

Fix: Added confusion matrix output and classification reports to all three classification models.

4 Model Performance Summary

Model	**Task**	**Metric**	**Value**	**Parameters**
DNN	Regression (Rating)	MAE	∼0.73	3,009
CNN (MobileNetV2)	Binary Classification	Accuracy	∼57%	2.4M (most frozen)
LSTM (Bidirectional)	3-Class Sentiment	Accuracy	∼100%*	361K
Transformer	3-Class Sentiment	Accuracy	∼93%	362K
Multi-Modal	Regression (Rating)	MAE	∼0.70	2.3M (most frozen)

Table 1: Model Performance Comparison (*Note: 100% accuracy on LSTM is due to dataset simplicity – retrain after dataset expansion)

5 Grade Improvement Roadmap

Based on the grading rubric (20 marks total):

Criterion	**Max**	**Est. After Fixes**
Data & Problem Understanding	3	2.0 – 2.5
Modeling & Architecture Design	5	3.5 – 4.0
Implementation & Technical Quality	2	1.5 – 2.0
Experimental & Comparative Analysis	2	1.5 – 2.0
Teamwork & Individual Contribution	2	1.5 – 2.0
Evaluation Metrics & Validation	1	0.5 – 0.8
Individual Understanding (Viva)	5	Depends on viva
Total (excluding viva)	**15**	**10.5 – 13.3**

Table 2: Estimated Grade After All Fixes

6 Instructions for Re-running

After pulling the fixes, execute the notebooks in order:

1. `01_Data_Scraping.ipynb` – Regenerate dataset with expanded reviews/images

2. `02_Data_Preprocessing.ipynb` – Preprocess with corrected features (no rating_scaled)
3. `03_Model_1_DNN.ipynb` – Train DNN with corrected 4 features
4. `04_Model_2_CNN.ipynb` – Train CNN with added confusion matrix
5. `05_Model_3_LSTM.ipynb` – Train LSTM with expanded reviews + confusion matrix
6. `06_Model_4_Transformer.ipynb` – Train Transformer with positional encoding + confusion matrix
7. `07_Model_5_MultiModal.ipynb` – Train Multi-Modal with corrected features
8. `08_Model_Comparison.ipynb` – Run comparative analysis

Conclusion

The project now has all critical issues resolved. The data leakage has been eliminated, the Transformer correctly implements positional encoding, the dataset has more variety, and a comprehensive comparative analysis is included. The estimated grade has improved from 5.8–9.0 to 10.5–13.3 (out of 15, excluding viva).